



Interior designer

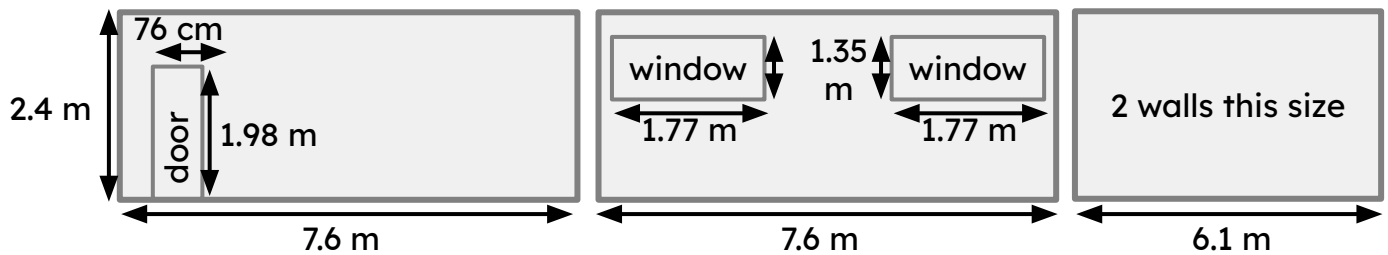
Task A: Measurements in interior design

1) Below are sketches of the walls in a room. (Not to scale)

A 2.5 litre tin of paint covers 25 m^2 .

The walls will need 2 coats of paint.

How many tins of paint are needed to paint the room?



$$\text{Total area of walls requiring painting} = 65.76 - 6.2838 = 59.4762 \text{ m}^2$$

$$\text{Total area requiring painting} = 118.9524 \text{ m}^2$$

$$\text{Number of tins required} = 118.9524 \div 25$$

$$= 4.758...$$

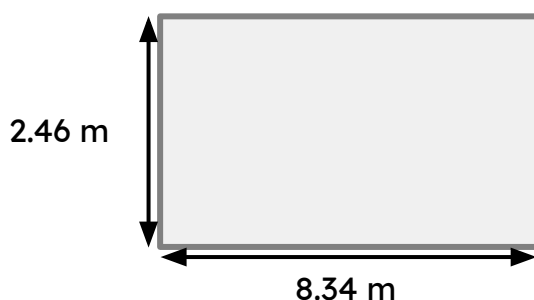
$$= 5 \text{ tins}$$

2) An interior designer wants to wallpaper a wall.

Below is a sketch of the wall.

The width of a roll of wallpaper is 52 cm and the length of the roll is 10.5 m.

How many rolls of wallpaper are needed to paper this wall?



$$8.34 \div 0.52 = 16.038...$$

17 lengths needed

$$10.5 \div 2.46 = 4.268...$$

4 lengths from each roll

$$17 \div 4 = 4.25$$

5 rolls are needed



Task B: Finance in interior design

1) The total area that needs painting in a house is 211.2 m^2 .

The paint covers 16 m^2 per litre.

The walls will need 2 coats of paint.

Calculate the cost to paint the room, ensuring that there is minimum wastage.

Size of tin (l)	Cost (£)
2.5	45.49
5	85.99
10	130.99

$$\begin{aligned}
 \text{Number of litres needed} &= 211.2 \times 2 \div 16 \\
 &= 26.4 \\
 &= 27 \text{ litres}
 \end{aligned}$$

The closest you can get to 27 litres is 2×10 litres, 1×5 litres and 1×2.5 litres.

$$\begin{aligned}
 \text{Cost} &= 2 \times £130.99 + £85.99 + £45.49 \\
 &= £393.46
 \end{aligned}$$

2) I need the following lengths of wood:

$4 \times 1.9 \text{ m}$, $6 \times 1.7 \text{ m}$ and $4 \times 0.5 \text{ m}$

a) What is the cost of the wood with minimum wastage?

b) How much wastage is there?

Length (m)	Price (£)
2.4	4.20
3	5.25
3.6	6.30
4.2	7.35
4.8	8.40

$4 \times 2.4 \text{ m}$ makes $4 \times 1.9 \text{ m}$ lengths and $4 \times 0.5 \text{ m}$ lengths left over.
 $3 \times 3.6 \text{ m}$ makes $6 \times 1.7 \text{ m}$ lengths.

$$a) 4 \times £4.20 + 3 \times £6.30 = £35.70$$

b) 0.6 m wastage

